

27 Statistical Estimation (From Probability to Statistics)

Example 27.1. In each of the following scenarios, one of (i) and (ii) is a *probability* question and one is a *statistics* question. Classify each question as “probability” or “statistics”. Then discuss some general features of the probability questions, and some general features of the statistics questions

1. U.S. Satisfaction

- Assume that 27% of Americans are satisfied with the way things are going in the U.S. If a random sample of 1000 Americans is selected, what are the chances that fewer than 250 Americans in the sample are satisfied with the way things are going in the U.S?
- In a random sample of 1000 Americans, 250 Americans are satisfied with the way things are going in the U.S. What percent of all Americans are satisfied with the way things are going in the U.S?

2. Great white shark length

- Assume that lengths of female great white sharks follow a Normal distribution with mean 16 ft and standard deviation 3 ft. If a random sample of 40 female great white sharks is selected, what are the chances that the average length in the sample is between 16.5 and 17.5 ft?
- In a sample of 40 female great white sharks, the sample mean length is 17.0 ft. What is the average length of female great white sharks?

3. In a [clinical trial](#), 200 subjects are randomly assigned to receive either an experimental vaccine or no treatment (e.g., a placebo), and then each subject is tested to see if they developed immunity.

- If the vaccine is in reality not effective, what are the chances that the proportion of subjects who develop immunity is at least twice as large for those who received the vaccine than for those who did not?

- Among the 100 subjects who received the vaccine, 20 developed immunity; among the 100 subjects who did not receive the vaccine, 10 developed immunity. Is there evidence that the vaccine is effective?

4. Snap streaks

- Assume that lengths of streaks on Snapchat follow an Exponential distribution, with mean 75 for teen users and mean 40 for adult users. A random sample of 200 Snapchat users is selected, 100 teens and 100 adults. What are the chances that the sample mean streak length for teens is within 20 of the sample mean streak length for adults?
- In a random sample of 100 teen Snapchat users, the sample mean streak length is 80. In a random sample of 100 adult Snapchat users, the sample mean streak length is 35. In general, how much longer do streak lengths for teen Snapchat users tend to be than for adult Snapchat users?

- **Probability:** Assume a model for a random (uncertain) process and evaluate probabilities of potential outcomes or events — what might the data look like?
 - In probability, a model is assumed and parameters are assumed to be known, but the data is unknown.
 - In probability, we typically call the model a “distribution” and the data “random variables”.
- **Statistical inference:** Observe sample data and make conclusions about the process that generated the data
 - In statistics, the data is observed but parameters are unknown.
 - In statistics, we typically call the model the “population” and the data the “sample”.
- *Statistical inference* involves using sample data to make conclusions about the population.
 - **Point estimation:** Estimate an unknown population parameter with a single number based on sample data

- **Interval estimation:** Estimate an unknown population parameter with a plausible range of values based on sample data
- **Hypothesis testing:** Assess the evidence that the sample data provides regarding a particular claim about unknown population parameters.
- The *population distribution* is the distribution of individual values of the variable of interest. A *parameter*, generically denoted θ , is a number that summarizes the population distribution.
- The **population distribution**, denoted p_θ or $p_\theta(x)$, is a probability model for the distribution of values of a variable¹ X over all *individuals* in the population.
 - If the variable X is discrete, then p_θ represents a probability mass function (pmf): $p_\theta(x) = P(X = x)$
 - If the variable X is continuous, then p_θ represents a probability density function (pdf): $p_\theta(x) \approx \frac{1}{\epsilon}P(x - \epsilon/2 < X < x + \epsilon/2)$
- A **parameter**, generically denoted θ , is a characteristic of the population distribution, often computed as an expected value of some function of X .
 - In some cases θ represents a vector of multiple parameters.
 - For example, if p_θ represents a Normal distribution, then θ might represent the pair $\theta \equiv (\mu, \sigma)$ consisting of the population mean and the population standard deviation.
 - The **population mean**, typically denoted μ , is the average of the values of the variable over all individuals in the population, $\mu = E(X)$
 - The **population standard deviation**, typically denoted σ , is the standard deviation of all the individual values in the population, $\sigma = SD(X)$
- A parameter is a *fixed number* but its value is *unknown*².
- Many statistical studies involve *random sampling* from a population.
- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (i.i.d.)

¹Recall: By definition $\text{Var}(Y) = E[(Y - E(Y))^2]$ but remember the useful computational formula

$\text{Var}(Y) = E(Y^2) - (E(Y))^2$. Apply this to $Y = \hat{\theta} - \theta$, and remember that $\hat{\theta}$ is random but θ is not.

²There are two main philosophical approaches to statistical inference that differ mainly in whether they emphasize “fixed number” or “unknown”. In our brief exploration of estimation, we will follow the *frequentist* approach, which treats a parameter as a *fixed number*. The *Bayesian* approach treats an *unknown* parameter—in other words, an *uncertain* parameter—as a *random variable* with a probability distribution that describes the degree of uncertainty. Both approaches are valid, each with advantages and disadvantages. The frequentist approach has been historically more prevalent, but the Bayesian approach has gained in popularity in the last 30 or so years.

- Identically distributed — each individual X_i has the same marginal distribution — the population distribution — since they are sampled from the same population:

$$X_i \sim p_\theta \quad \text{for all } i = 1, \dots, n$$

- Independent — the values are sampled independently, so the joint distribution³ is the product of marginal distributions:

$$p_\theta(x_1, \dots, x_n) = p_\theta(x_1)p_\theta(x_2) \cdots p_\theta(x_n) \quad \text{for all } x_1, \dots, x_n$$

- A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ .
 - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
 - The probability distribution of a statistic is called a **sampling distribution**.
 - The *distribution* of a statistic will depend on θ , but the *value* of a statistic will not.
 - The **sample mean**, typically denoted $\bar{X}$, is the average of the values of the variable over all individuals in the population, $\bar{X} = (X_1 + \cdots + X_n)/n$
- A statistic $\hat{\theta} \equiv T(X_1, \dots, X_n)$ used to estimate a parameter θ is called a **(point) estimator**.
 - Naturally, we hope that there is some correspondence between our estimator $\hat{\theta}$ and the parameter it is estimating θ — for example, estimating the population mean with the sample mean — but the definition does not require this.
 - When the numerical value of a statistic is calculated from observed sample data, its value is called a *(point) estimate* of the parameter.
 - The estimator $\hat{\theta}$ is a RV whose value changes from sample to sample; an estimate is the observed value computed based on the results of a single sample.

Example 27.2. Jerry has just opened a car dealership in a new location. He assumes that the number of cars he'll sell in any randomly selected day has a $\text{Poisson}(\mu)$ distribution. But he doesn't know the value of μ , so he plans to use data from n days of business to estimate μ . Let $X_1, \dots, X_n$ represent the number of cars sold for a sample of n days.

³Abuse of notation in the expression: p_θ on the left represents the joint distribution of $(X_1, \dots, X_n)$, a function with n inputs; p_θ on the right represents that marginal distribution of each X_i (since i.d.), a function with a single input.

1. What plays the role of the population distribution? What is the parameter? How might you interpret the parameter in this context?

2. Suggest one estimator of μ . (Hint: what does μ represent for a Poisson distribution?)

3. Suppose he collects data on $n = 3$ days and observes $x_1 = 3$ cars sold on day 1, $x_2 = 0$ cars sold on day 2, and $x_3 = 2$ cars sold on day 3. Compute the value of your estimator from the previous part for this sample. That is, what is your estimate of μ based on this sample?

4. Suggest some other estimators of μ . (Hint: what else does μ represent for a Poisson distribution?)

5. For a $\text{Poisson}(\mu)$ distribution μ is both the population mean and the population variance. So one reasonable estimator of μ is the sample mean $\bar{X}$. Another reasonable estimator is the sample variance. But there are two commonly used formulas for the sample variance, each of which provides an estimator of μ

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Compute the value of each of these estimators for the $(3, 0, 2)$ sample. That is, what is your estimate of μ based on this sample if you use each of these estimators?

6. At his previous dealership at a different location, Jerry sold 2.3 cars per day on average. Consider the estimator of μ : $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$. Explain in words what this estimator represents. In particular, what happens as n increases? Then compute the value of this estimator for the (3, 0, 2) sample.

7. According to the Poisson(μ) distribution, the probability of selling 0 cars in a day is $p_0 = e^{-\mu}$. Rearranging, $\mu = -\log p_0$. Therefore, if $\hat{p}_0$ is the sample proportion of days with 0 cars sold, then another estimator of μ is $-\log \hat{p}_0$. Compute the value of this estimator for the (3, 0, 2) sample.

8. We have seen a few different, seemingly reasonable estimators of μ , and each produced a different estimate of μ based on the same (3, 0, 2) sample data. Which of these numbers is the best *estimate* of μ ? That is, which of these estimates is closest to μ ?

9. How might you investigate which of the *estimators* corresponds to the best estimation *procedure*?
 - Because the true value of the parameter θ is unknown, we never know if the sample data provides a reasonable estimate of θ
 - Rather, we evaluate the estimation *procedure* for potential values of θ : How does the estimation procedure perform over many possible samples given a range of potential values of θ ?
 - Some questions of interest when estimating a parameter θ

- How do we find an estimator of θ ?
- What are properties of “good” estimators?
- How can we tell if one estimator is “better” than another?
- Can we find a “best” estimator of θ ? (Short answer: no)
- How do we determine the margin of error for an estimator?